Kushagra Srivastava

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Education

Indian Institute of Information Technology, Allahabad **Aug 2024 - Dec 2026**
Master of Technology in Information Technology (Robotics & Machine Intelligence) (CGPA: 8.83) *Prayagraj, Uttar Pradesh*

KIET Group of Institutions, APJ Abdul Kalam Technical University **Nov 2020 – Jun 2024**
Bachelor of Technology in Computer Science & Engineering (CGPA: 8.46) *Ghaziabad, Uttar Pradesh*

Experience

Software Engineering Intern **Aug 2023 – Aug 2024**
Simplify Workforce Private Limited *Noida, Uttar Pradesh*

- Prototyped LLM-powered enterprise chat interfaces as a proof-of-concept (POC), evaluating multi-server (upto 5 servers) deployment patterns and load-balancing strategies for high-query throughput systems.
- Built 5+ ETL pipelines integrating MySQL with Databricks Lakehouse to support PowerBI-driven OLAP analytics, and architected SQL-based data models with production-grade views for client-specific slicing across UAT, pre-production, and production environments.
- Owned 10+ reporting workflows within a multi-tenant analytics platform, collaborating with Product Owners to convert evolving client requirements into data models and analytical logic, driving client-specific customization and supporting enterprise onboarding transitions.

Student Software Developer, Research & Development Team **Feb 2022 – Oct 2022**
MBS India Pvt. Ltd. *Ghaziabad, Uttar Pradesh*

- Constructed a Python (Qt v5) real-time interface for edge-computing biomedical instrumentation, implementing deterministic serial communication layers to guarantee low-latency data exchange with ventilator hardware.
- Modeled a high-frequency telemetry logging ecosystem with ACID-compliant storage, performing cross-platform validation across 3 operating systems on Raspberry Pi nodes to ensure medical-grade robustness and system reliability.

Projects

Mujoco-Copilot — Agentic AI-Powered Design Studio | *Python, FastAPI, MuJoCo, Ollama* **Mar 2026**

- Architected a GenAI-driven MuJoCo simulation co-pilot, developing a high-throughput FastAPI orchestration layer to translate zero-shot natural language prompts into syntactically valid robot XML, integrated with a real-time simulation loop for immediate validation.
- Designed a multi-turn LLM self-correction loop with structured XML parsing and patch-based validation, enabling deterministic recovery from invalid generations via bounded retries (up to 3 retries) and configurable timeout control.
- Implemented a provider-agnostic execution layer with state-isolated workflows and JSONL-based telemetry, enabling traceable model interactions, reproducible runs, and production-grade observability.

Regolith — Cross-Terrain RL Robustness Evaluation Framework | *Reinforcement Learning, MuJoCo* **Feb 2026**

- Engineered a modular RL simulation stack supporting 3 terrain families with 256×256 heightfields and vectorized environments (8 envs), enabling scalable multi-terrain experimentation across 9+ seeded runs.
- Instrumented a physics-grounded analytics pipeline extracting 6+ terrain and contact-level features from 256×256 heightfields per step, enabling robustness evaluation and principled reward shaping for terrain-aware RL policies.
- Executed large-scale PPO training (100M+ steps) with vectorized environments, seed-controlled experimentation, and structured logging, enabling reproducible cross-terrain generalization studies and dataset-like experiment tracking.

Achievements

- **Leadership:** Operations Head & RL Team Lead at Robita (IIITA); Co-Founder, Technocrats (KIET); ML Lead, DSC KIET; Domain Head, MYCIN Innovation Center (KIET); Head Volunteer for International Workshops (2025 & 2026).
- **Programs:** Selected for Data Science for Social Good Fellowship (Germany, 2023); completed Google ML Bootcamp (2022) and AWS AI & ML Scholarship (2023); facilitated Google Crowdsource (2022) & Google Cloud Arcade (2023).
- **Teaching Assistant:** Assisted in delivering 4 undergraduate courses—Machine Learning, Data Structures & Algorithms, Programming for Problem Solving, and Design & Analysis of Algorithms—supporting course logistics, doubt resolution, and student evaluations.

Publications

Srivastava, Kushagra (Nov. 2022). **"Comparative Analysis of various Machine Learning Techniques to predict Breast Cancer"**. In: IEEE ICICT 2022 (DOI: 10.1109/ICICT55121.2022.10064517)

Srivastava, Kushagra (March 2024). **"Comparative Analysis of Deep Learning Models for Facial Emotional Recognition"**. In: IEEE ICRITO 2024 (DOI: 10.1109/ICRITO61523.2024.10522163)